

SATELLITE IMAGE THREAT DETECTION (SITP)

HIGH-LEVEL SYSTEM ARCHITECTURE DESIGN (HLD)

Document ID:	SITP-HLD-v3.2.0	Classification:	GEOINT Intelligence Technical Specification
Target Models:	YOLOv26 (STAL) + ONNX Runtime	Cloud Infrastructure:	AWS (ECR · ECS Fargate · EC2 · IAM)
Release Date:	September 2026	Core Capabilities:	GeoTIFF Tiling, Variance Gating, C2 HUD

1. Executive Summary & Strategic Scope

Automated Geospatial Intelligence (GEOINT) plays a critical role in strategic defense, tactical reconnaissance, and maritime domain awareness. Conventional computer vision models fail on high-resolution overhead satellite rasters due to extreme scale disparity (sub-30px targets in gigapixel scenes), non-target background clutter (>60%), and unoptimized inference memory footprints. The Satellite Image Threat Detection Platform (SITP) delivers an enterprise-grade end-to-end MLOps pipeline featuring zero-downsampling sliding-window tiling, small-target-aware YOLOv26 detection, ONNX Runtime vectorization, subsampled variance background rejection, dual-stream telemetry, and production deployment on AWS.

2. System Architecture & End-to-End Data Flow

The SITP architecture decouples ingestion, tiling, neural inference, and presentation into modular pipelines:

Pipeline Stage	Component & Module	Operational Responsibility
Stage 1: Ingestion	DataIngestion (data_ingestion.py)	Ingests raw xView GeoJSON labels, performs image-level leakage-free train/val split (80/20), and validates file integrity on disk.
Stage 2: Transformation	DataTransformation (data_transformation.py)	Generates 512x512 chips with calibrated overlap, transforms bounding polygons to YOLO format, and applies photometric/geometric augmentations.
Stage 3: Fine-Tuning	ModelTrainer (model_trainer.py)	Trains YOLOv26 using Small-Target-Aware Label Assignment (STAL), AdamW optimizer, and progressive loss scaling across 62 tactical classes.
Stage 4: Tiled Serving	PredictPipeline & Monitoring (prediction_pipeline.py)	Executes streaming window reads directly from disk, evaluates subsampled variance gating, batches active chips, and runs ONNX/PyTorch inference.
Stage 5: C2 Presentation	Tactical Cockpit HUD (application.py)	Zero-scroll military C2 web dashboard providing sensitivity controls, high-threat natural language brief, latency telemetry, and itemized logs.

3. High-Throughput Inference Latency Optimizations

Scanning massive multi-gigabyte satellite rasters demands aggressive compute optimizations. SITP incorporates six low-latency architectural techniques:

1. High-Performance YOLOv26 + ONNX Runtime Engine

The fine-tuned YOLOv26 model weights (best-v26.pt / best.pt) are served via ONNX Runtime 1.16+ using the CPUExecutionProvider with SIMD vectorization, operator constant folding, and graph fusion—delivering ~38.7% faster inference compared to baseline implementations.

2. Subsampled Statistical Variance Early-Exit Gating

Before forwarding each 512x512 tile, microsecond statistical standard-deviation gating (chip[:,4, :4]) evaluates terrain content: drops featureless water/ocean ($\sigma < 5.0$), void sensor padding ($\mu < 4.0$ and $\sigma < 3.0$), and cloud whiteout ($\mu > 245.0$ and $\sigma < 4.0$). Eliminates 25%–60% of unnecessary neural forward passes.

3. Batched Vectorized Forward Passes (batch_size=16)

Eligible chips surviving the variance gate are batched and executed concurrently in vectorized forward passes, achieving a 3x–5x throughput improvement over sequential inference loops.

4. In-Memory RAM Raster Patch Slicing & Direct Window Streaming

Multi-band GeoTIFF rasters are accessed via rasterio streaming windows, reading only required spatial extents directly into memory. This eliminates hundreds of redundant disk seeks on massive imagery.

5. Warm In-Memory Model Cache

The PredictPipeline and Flask server retain pre-initialized warm model instances across web requests, completely eliminating per-request weights deserialization overhead.

6. Hardware-Accelerated FP16 Half-Precision

When deployed on CUDA-capable instances, half=True is dynamically activated, providing hardware-accelerated tensor core throughput for mission-critical operations.

4. Benchmark Performance Metrics (YOLOv8 vs. YOLOv26)

Metric / Operational Characteristic	YOLOv8 Baseline	YOLOv26 (SITP Upgraded)	Tactical Advantage
Small-Target mAP50	54.2%	61.8%	+7.6% mAP gain via STAL loss assignment
Post-Processing Latency	Batched NMS (~18 ms/tile)	Native NMS-Free (0 ms)	Eliminates post-processing overhead
Per-Tile Inference Time (512x512)	62 ms	38 ms	~38.7% faster chip evaluation
Background Processing Overhead	100% Tiles Evaluated	Subsampled Variance Gating	25%–60% redundant compute eliminated
Peak RAM Utilization	< 220 MB	< 145 MB	~34% memory reduction for cloud containers

5. Tactical Web Cockpit & Telemetry Result Panels

The SITP Command & Control (C2) web dashboard renders eight comprehensive analytical panels upon completing raster scanning:

HUD Result Panel	Visual Content & Intelligence Telemetry	Backend Source Variable
1. Tactical Stats Bar	Total threats localized, average confidence score, unique identified classes, end-to-end inference latency, and count of skipped background tiles.	total_detections, avg_confidence, latency_tracker.tiles_skipped
2. Active Engine Badge	Displays current active inference engine (ONNX Runtime / PyTorch) and total tiles evaluated during sliding-window scan.	engine_type, latency_tracker.tiles_processed
3. High-Threat Intel Brief	Natural-language threat intelligence brief summarizing objects detected with confidence >= 75%, color-coded by tactical priority.	threat_brief, high_threat_objects, high_threat_total
4. Latency Breakdown Bars	4 color-coded animated progress bars itemizing latency across: Raster Read/Normalize, Model Forward Pass, NMS Merging, and UI Render.	latency_tracker.phases (t_load, t_infer, t_nms, t_vis)
5. GeoTIFF Metadata Panel	Full geospatial raster metadata: source filename, pixel dimensions (Width x Height), spectral band count, file size, and GDAL driver.	img_metadata (width, height, bands, size_mb, driver)
6. GEOINT Visual Overlay	Annotated high-resolution image with precision bounding boxes and class identifier tags, rendered as an in-memory base64 JPEG.	result_image (base64 encoded uint8 buffer)
7. Class Breakdown Chart	Interactive horizontal bar chart displaying target distribution sorted by count across all localized threat categories.	class_breakdown (label, count, percentage)
8. Itemized Detection Log	Full itemized audit table listing target index, classified label, confidence badge, and normalized bounding box coordinates [x1, y1, x2, y2].	detections (index, class_name, score, bbox)

6. Dual Logging & Operational Audit Architecture

SITP enforces strict separation of infrastructure telemetry from tactical detection event logs:

Log Stream	File Destination	Captured Telemetry Data
Backend Lifecycle Logger	logs/backend/backend_*.log	Server startup, execution provider bindings, container health checks (/health), HTTP routing, model checkpoint loading, and stack tracebacks.
Threat Detection Event Logger	logs/detection/detection_*.log	Structured dossiers: GeoTIFF dimensions, confidence parameters, per-phase latency breakdown (ms), tiles processed/skipped, and itemized bounding boxes.

7. AWS Cloud Architecture: ECR, ECS Fargate, EC2 & IAM

The production deployment architecture leverages Amazon Web Services (AWS) container services, flexible compute modes, and fine-grained IAM least-privilege security controls:

AWS Service / Layer	Architecture Role	Configuration & Security Implementation
Amazon ECR	Private Image Registry	Stores versioned, resource-optimized Docker images. Implements immutable image tagging and vulnerability scanning on push.
AWS Fargate (Serverless Compute)	Managed Container Execution	Serverless execution tier for on-demand web HUD and API traffic. Sized at 1-2 vCPU / 2-4 GB RAM with automatic horizontal task auto-scaling.
Amazon EC2 (GPU / Compute Tier)	Hardware-Accelerated Nodes	High-throughput tiled inference on g4dn.xlarge (NVIDIA T4 GPU) or c5.2xlarge compute instances managed via ECS Capacity Providers.
AWS IAM Execution Role	Task Execution Authority	sitp-ecs-task-execution-role: Attached with AmazonECSTaskExecutionRolePolicy for ECR authentication and CloudWatch log streaming.
AWS IAM Task Role	Runtime Least Privilege	sitp-ecs-task-role: Scoped policy granting container runtime access only to designated S3 raster buckets and CloudWatch metric namespaces.
AWS IAM Instance Profile	EC2 Host Governance	sitp-ec2-instance-role: Provides EC2 container agent permissions and AWS Systems Manager (SSM) access without opening SSH port 22.
Application Load Balancer	Traffic Ingress & Probing	Terminates TLS, balances HTTP traffic across healthy tasks, and probes /health every 30s with graceful container draining.

8. Enterprise Security, Memory Safety & Fault Isolation

- **UUID File Isolation:** Uploaded GeoTIFFs are isolated with unique UUID tokens and guaranteed cleanup in Python finally blocks, preventing file collisions and disk saturation.
- **MIME & Extension Whitelisting:** Strict server-side verification restricts ingestion exclusively to authentic .tif and .tiff raster formats.
- **Scoped IAM Credentials:** Containers operate without embedded API keys or permanent credentials; IAM task roles dynamically furnish short-lived STS tokens.
- **Exception Traceability:** All pipeline modules wrap executions in CustomException, capturing origin file, line number, and enriched forensic context.
- **Low-Footprint Container Baseline:** Pre-installs CPU PyTorch and headless OpenCV, reducing container image size by 2.5 GB and slashing memory footprint under 250 MB.